

Linear Algebra Midterm Examination

Question 1 (20 points)

Let A be the matrix:

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & -2 \\ 3 & 1 & 1 \end{bmatrix}$$

- Find the determinant of A .
- Find A^{-1} (the inverse of A).
- Solve the system $Ax = b$ where $b = [1, 0, 2]^T$.

Question 2 (25 points)

Let $V = \{[x_1, x_2, x_3]^T \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 0\}$.

- Show that V is a subspace of $\mathbb{R}^3$.
- Find a basis for V .
- What is the dimension of V ?
- Find the orthogonal complement $V^\perp$.

Question 3 (25 points)

Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by:

$$T([x_1, x_2, x_3]^T) = [2x_1 - x_2 + x_3, x_1 + 3x_2]^T$$

- Find the standard matrix representation of T .
- Find the kernel of T (null space).
- Find the range of T (column space).
- Determine if T is one-to-one and/or onto.

Question 4 (20 points)

Find the eigenvalues and eigenvectors of the matrix:

$$B = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$$

Is B diagonalizable? If so, find a diagonal matrix D and an invertible matrix P such that $B = PDP^{-1}$.

Question 5 (10 points)

Let $u = [1, 2, -1]^T$ and $v = [3, 0, 1]^T$.

- Compute $u \cdot v$ (dot product).
- Compute $\|u\|$ and $\|v\|$ (norms).
- Find the angle between u and v.
- Compute $u \times v$ (cross product).

Useful Formulas:

- $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$
- For 3×3 matrix, $\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$
- $u \cdot v = \|u\| \|v\| \cos(\theta)$
- Eigenvalues satisfy $\det(A - \lambda I) = 0$